

Tanmay Debnath

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Ph.D. Student

The Roslin Institute ([Roslin](#))

The University of Edinburgh, Easter Bush Campus, Midlothian EH25 9RG

PRINCIPAL INTERESTS

Machine learning, Reinforcement Learning (simulation), Bioinformatics, Image processing, Image based artificial intelligence analysis, Feature simulation, Technical Writing, Biomedical Engineering

RESEARCH EXPERIENCE

Research Assistant

October 2021 - September 2023

Toyohashi University of Technology

Toyohashi, Japan

- Working in the Mechanical Engineering department, under the guidance of [Prof. Moeto Nagai](#) as a graduate researcher
- Developed a new object detection model, designed specially for cells at high and low magnification, for real-time detection of cells and cell viability prediction. Achieved an mAP value greater than 0.94
- Working on multi-armed bandit problem to find the parameters important for optimization of the optical system and photocuring process, for single-cell extraction. Optimized intensity using RL
- Worked on explainable AI systems

AI Engineer

August 2022 - September 2022

AWL Inc.

Hokkaido, Japan

- Worked as an artificial intelligence engineer in the Research and Development division of the company
- Developed a core feature (model fine-tuning pipeline) of their current product which is integrated with its beta version
- Researched and implemented different AI architectures for feature extraction and clustering of the company-generated data

ML Engineer/Research Assistant

July 2021 - December 2021

Case Western Reserve University

Cleveland, US

- Worked in the Biomedical Engineering department, under the guidance of [Prof. Satish E. Viswanath](#) as a deep learning researcher
- developed and tested a Quality Assessment model (based on *ResNet-18*), using PyTorch, in ABIDE dataset, reaching 94% and 91% in training and validation accuracies
- Prepared SHAP and GRAD-CAM visualizations for explainable AI systems to characterize model learning.
- Research on Self-supervised learning for extracting features from radiological images autonomously

EDUCATION

The University of Edinburgh, Edinburgh, UK

October 2023 – September 2027

The Roslin Institute

PhD in Genetics and Genomics

- Research in Bioinformatics at the Roslin Institute, under the direction of [Prof. James Prendergast](#).
Focus areas: Bioinformatics, Comparative genomics, ML model development
Research Focus: Discovering regulatory grammar in non-model mammals using sequence-based deep learning models

Toyohashi University of Technology, Toyohashi, Japan

October 2021 – September 2023

Master of Engineering in Mechanical Engineering

GPA: 3.94/4.00

- Research in Bio-MEMS, under the direction of [Prof. Moeto Nagai](#).
Focus areas: Optoporation, Artificial Intelligence, Single-cell analysis
Dissertation: Implementation of machine learning tools and technologies for the automation of various BioMEMS processes

- **Focus areas:** medical imaging and deep learning.

Thesis: Mathematical formulation/analysis of polyp detection and segmentation and deployment of advanced image processing/computer vision and machine learning techniques in the video-colonoscopy of the gastro-intestinal tract

POSITIONS OF RESPONSIBILITY

- Aug 2024 - Present Treasurer of the Easter Bush Society
- Sep 2024 - Present Division Representative of the Quantitative Biology Division, Roslin Institute

ACHIEVEMENTS

Roslin Foundation scholarship | *Fully-Funded* 2023 – Present

- PhD in Edinburgh, covering all expenses in 4 years

MSCA DC scholarship | *Fully-Funded; Offer Declined* 2023

- Received the prestigious Marie Skłodowska-Curie Actions DC scholarship for conducting research on compressed sensing using reinforcement learning methods at the University of Eastern Finland in the Department of Physics

MEXT scholarship | *Fully-Funded* 2021 – 2023

- Master of Engineering in Japan, worth 41000 USD, covering all expenses in 2 years
- Expenses mainly include full tuition fee waiver and monthly stipend

JICA Scholarship | *Fully-Funded* 2021 – 2023

- Scholarship mainly bearing the health insurance, research grants and academic conference travel grants

GLOBAL Recognitions | *Awards* 2020

- MIT Hacking Medicine 2020 Global winner
- MIT COVID-19 hackathon 2020 Global finalist
- Academic grant for *Indian representative* to Beijing, China in July 2019

CONFERENCES, JOURNALS AND REPORTS

Journals

1. **T. Debnath**, et al. Deep learning can automate chicken tibia-breaking strength quantification to improve animal welfare. *Poultry Science* Vol. 105, Issue 4, 106549 (2026). DOI: 10.1016/j.psj.2026.106549
2. **T. Debnath**, R. Hattori, S. Okamoto et al. Automated detection of patterned single-cells within hydrogel using deep learning. *Sci Rep* 12, 18343 (2022). DOI: 10.1038/s41598-022-22774-0

Conferences

3. **T. Debnath**, P. Wilson, R. P. Wong, L. Plenderleith, B. Andersson, M. Schmutz, I. Dunn and J. Prendergast, "Deep learning to automate chicken tibia-breaking strength annotation from X-ray images to improve animal welfare". *ESPG 2025; Gdansk, Poland (Accepted/ Oral Presentation)*
2. **T. Debnath**, L. Plenderleith, L. Morrison, M. Hassan and J. Prendergast, "Understanding regulatory grammar in non-model mammals using deep learning". *ISMB 2025; Liverpool, UK (Accepted/ Poster Presentation)*
1. **T. Debnath**, R. Hattori, S. Okamoto, T. Shibata and M. Nagai, "Automated Detection of Patterned Single-Cells using Deep Learning". *APCOT 2022; China (Accepted/ Oral Presentation)*

TECHNICAL AND EXPERIMENTAL SKILLS

Languages: Python, C++, Rust (basic)

Frameworks: PyTorch, TensorFlow, Fast.ai

Developer Tools: Git, Docker, VS Code, Visual Studio

Data Science Libraries: OpenCV, Scikit-Learn, SciPy, pandas, NumPy, Matplotlib, Albumentations

Scientific Illustrations: Adobe Illustrator

Experimentation: Cell culturing, Photoporation, Photocuring